General Chemistry I

Chemistry: The central science

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Lectures 7,8: Periodic properties of elements

Content

1. Development of the periodic table
2. Effective nuclear charge
3. Sizes of Atoms and Ions
4. Ionization energy
5. Electron affinity
6. Elements properties

Discovery of the elements

Ancient time

S

Au

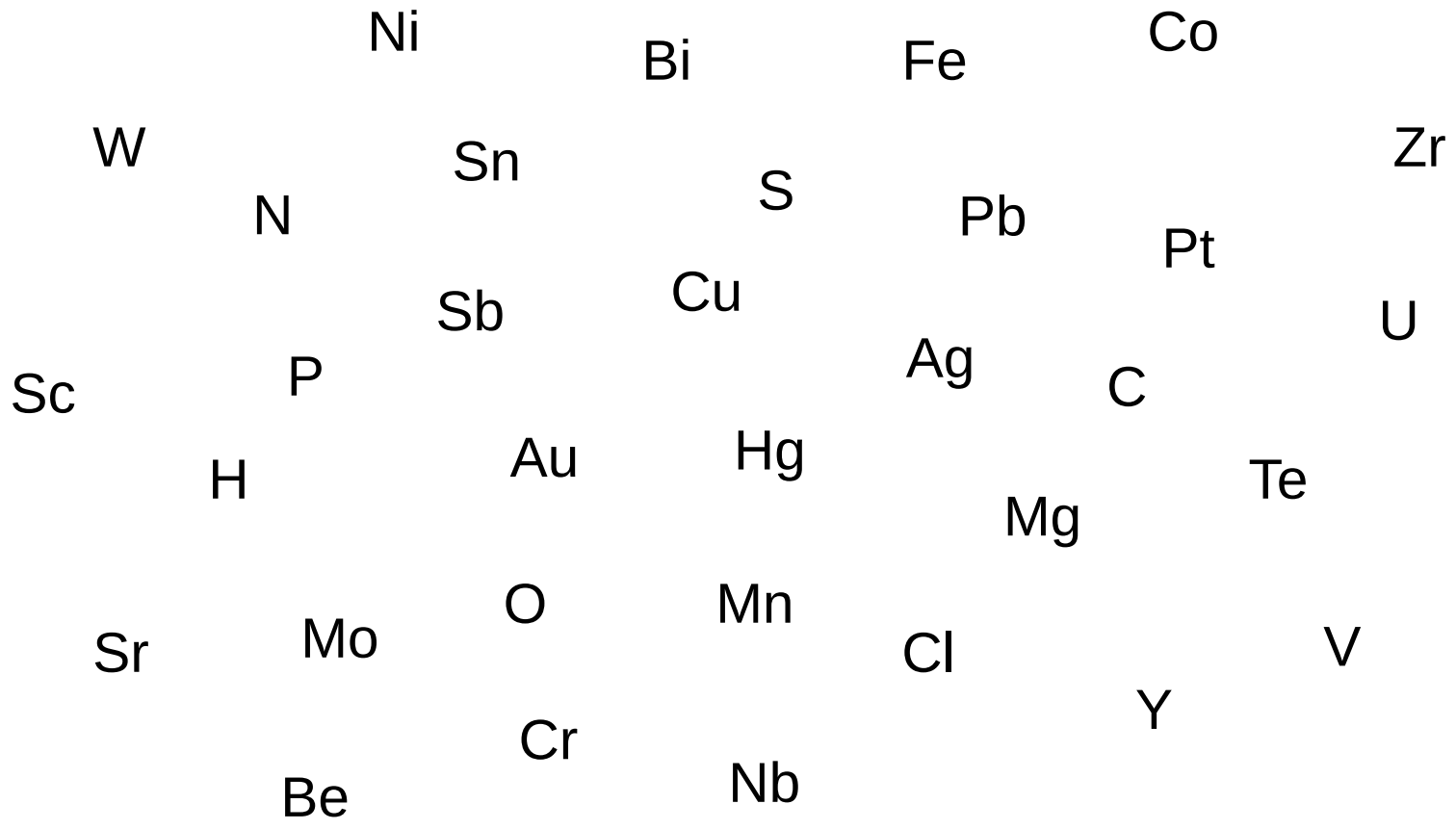
Cu

Ag



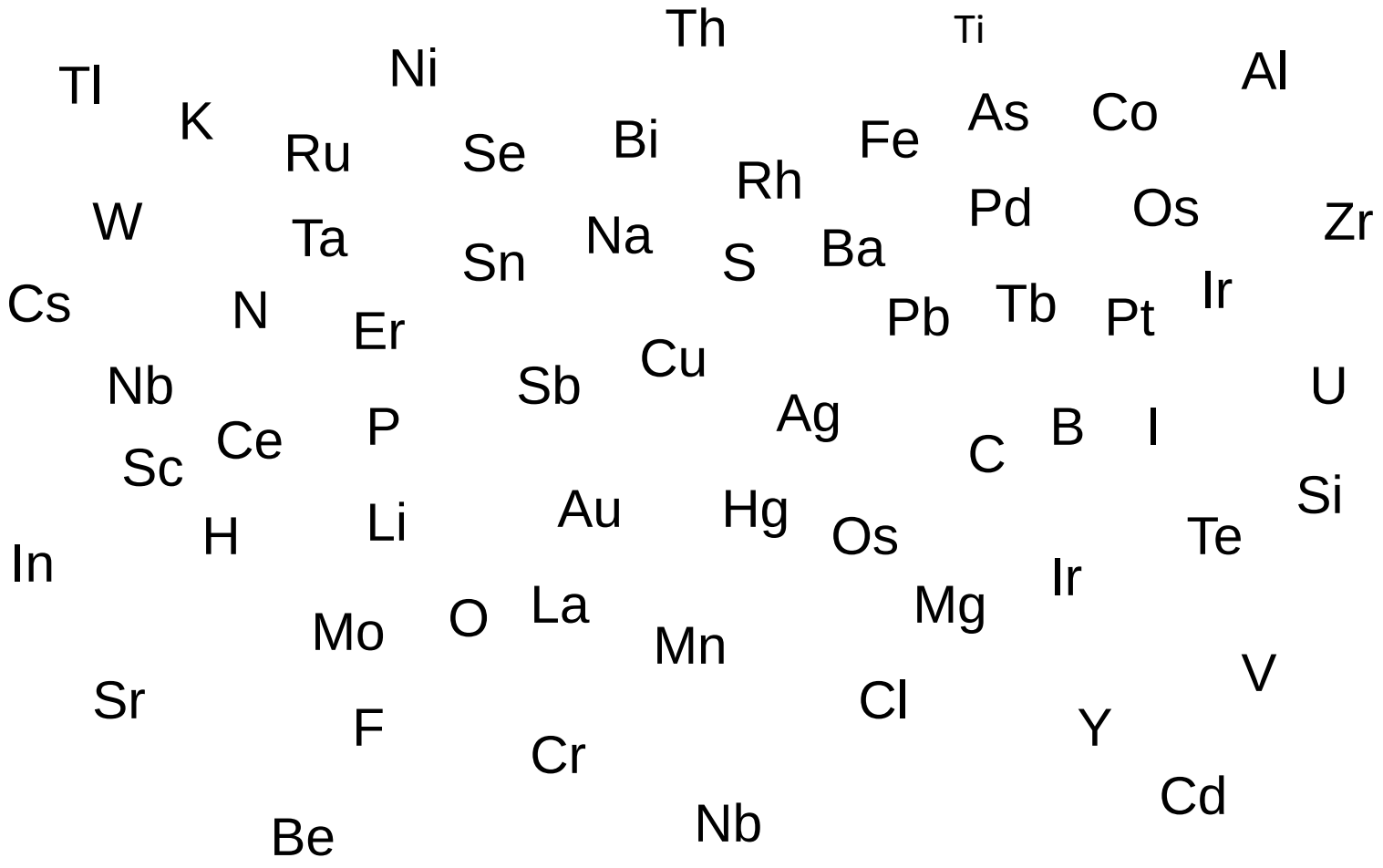
Discovery of the elements

By 1800: 33 elements



Discovery of the elements

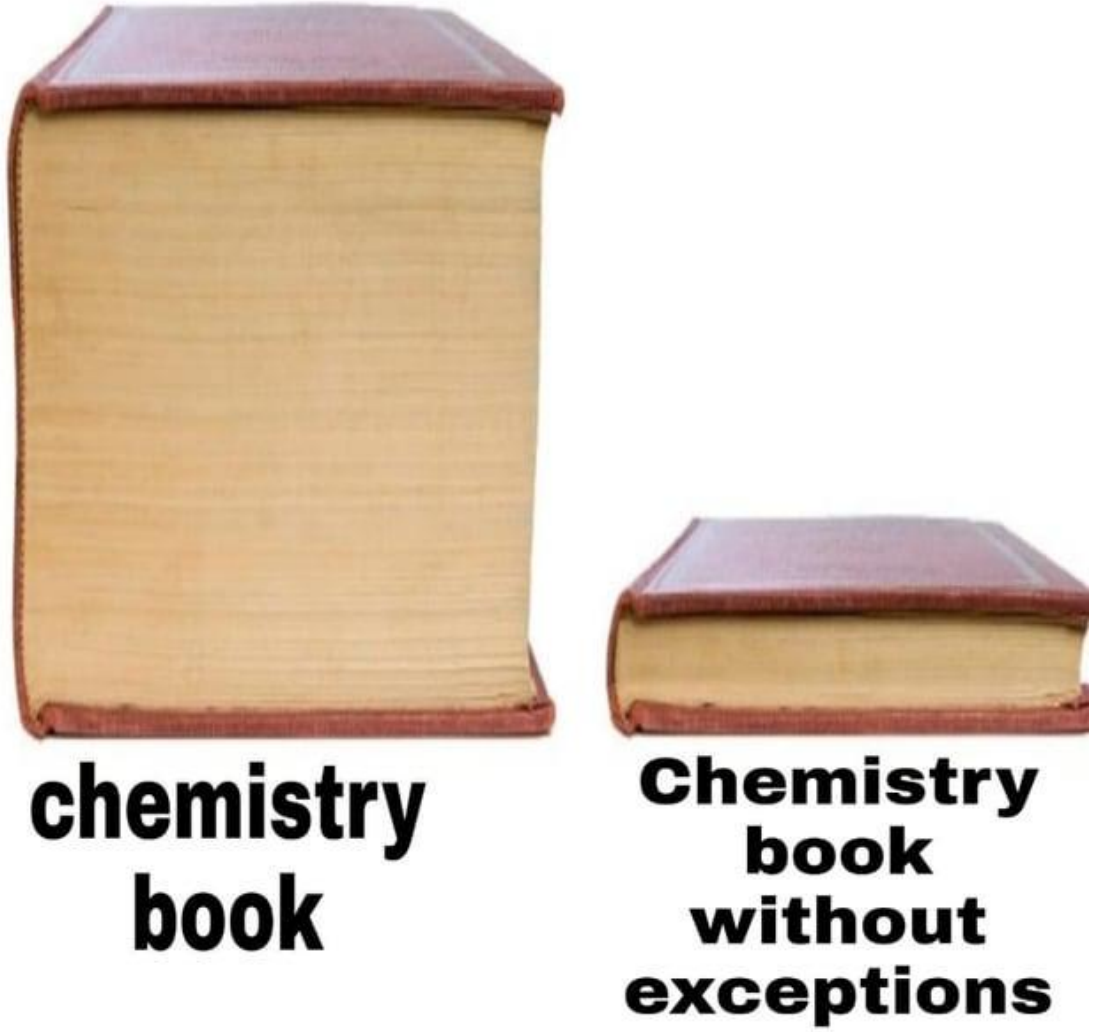
By 1865: 63 elements



I hate Chemistry because of EXCEPTIONS….?

By 1865: 63 elements

Shall we remember physical & chemical properties of all 63 elements (63 EXCEPTIONS)?



Development of the periodic table

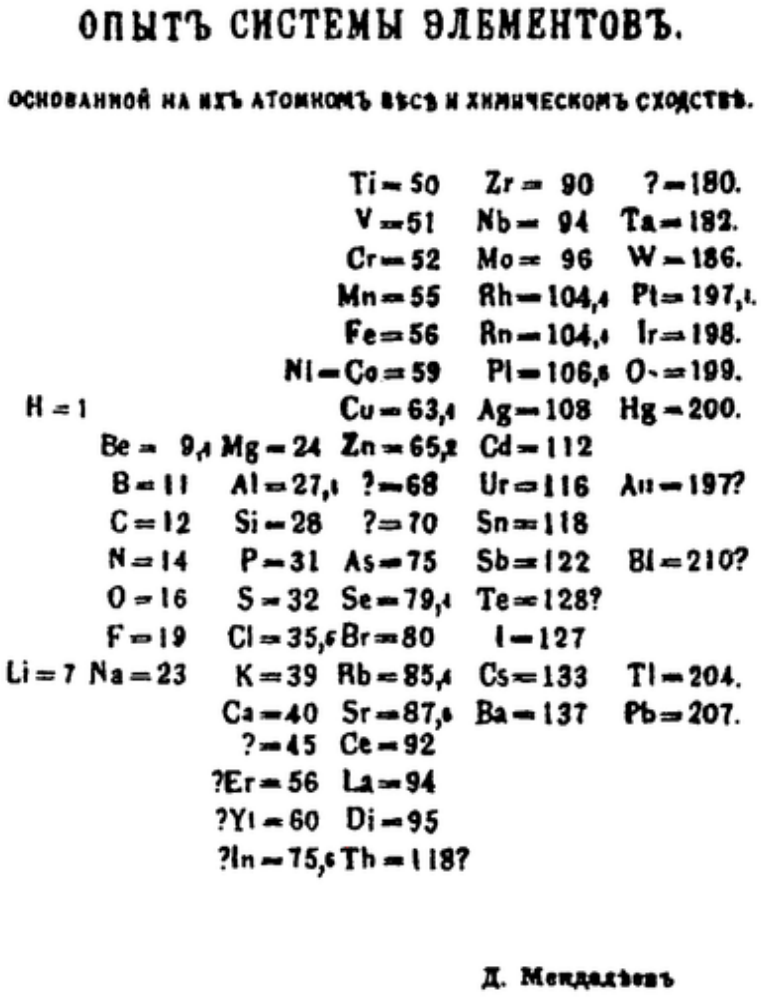
1869 - Dmitri Mendeleev (Russia) & Lothar Meyer (Germany) - independently proposed idea of periodicity (recurrence of properties)

Similar chemical and physical properties repeat periodically when the elements are arranged in order of increasing **atomic mass**

Periodic Table

Name chemical element	Symbol	Atomic mass
Hydrogen	H	1
Lithium	Li	6.941
Beryllium	Be	9.012
Boron	B	10.811
Carbon	C	12.01
Nitrogen	N	14
Oxygen	O	15.9994
Sodium	Na	22.989
Magnesium	Mg	24.305
Aluminum	Al	26.581
Silicon	Si	28.0855
Phosphorus	P	30.973
Sulfur	S	32.066

increasing **atomic mass**



ОПЫТЪ СИСТЕМЫ ЭЛЕМЕНТОВЪ.

ОСНОВАННОЙ НА ИХЪ АТОМНОМЪ ВѢСѢ И ХИМИЧЕСКОМЪ СХОДСТВѢ.

			Ti = 50	Zr = 90	? = 180.	
			V = 51	Nb = 94	Ta = 182.	
			Cr = 52	Mo = 96	W = 186.	
			Mn = 55	Rh = 104,4	Pt = 197,1.	
			Fe = 56	Ru = 104,4	Ir = 198.	
			Ni = Co = 59	Pl = 106,6	Os = 199.	
H = 1			Cu = 63,4	Ag = 108	Hg = 200.	
	Be = 9,4	Mg = 24	Zn = 65,2	Cd = 112		
	B = 11	Al = 27,4	? = 68	Ur = 116	Au = 197?	
	C = 12	Si = 28	? = 70	Sn = 118		
	N = 14	P = 31	As = 75	Sb = 122	Bi = 210?	
	O = 16	S = 32	Se = 79,4	Te = 128?		
	F = 19	Cl = 35,6	Br = 80	I = 127		
Li = 7	Na = 23	K = 39	Rb = 85,4	Cs = 133	Tl = 204.	
		Ca = 40	Sr = 87,6	Ba = 137	Pb = 207.	
		? = 45	Ce = 92			
		?Er = 56	La = 94			
		?Yt = 60	Di = 95			
		?In = 75,6	Th = 118?			

Д. Менделѣевъ

Mendeleev's Table

H														
Li	Be									B	C	N	O	F
Na	Mg									Al	Si	P	S	Cl
K	Sc	Ti	V	Cr	Mn	Fe	Ni	Co	Cu			As	Se	Br
Rb	Sr	Y	Zr	Nb	Mo									

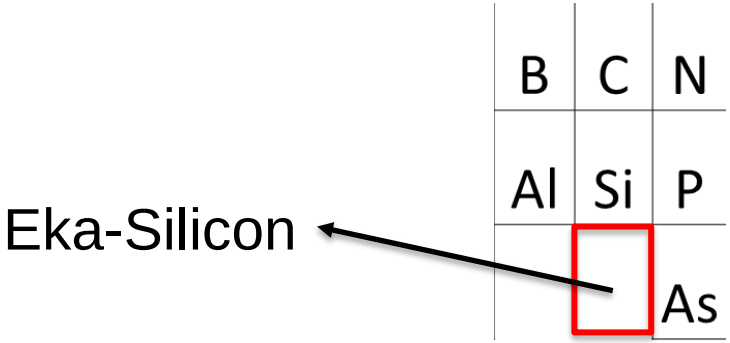
Development of the periodic table

- The tables of elements advanced by Mendeleev and Meyer were the forerunners of the modern periodic table
- **Mendeleev**
  - Grouped elements according to properties
  - Predicted properties for elements not yet discovered (*e.g.* Ga and Ge)

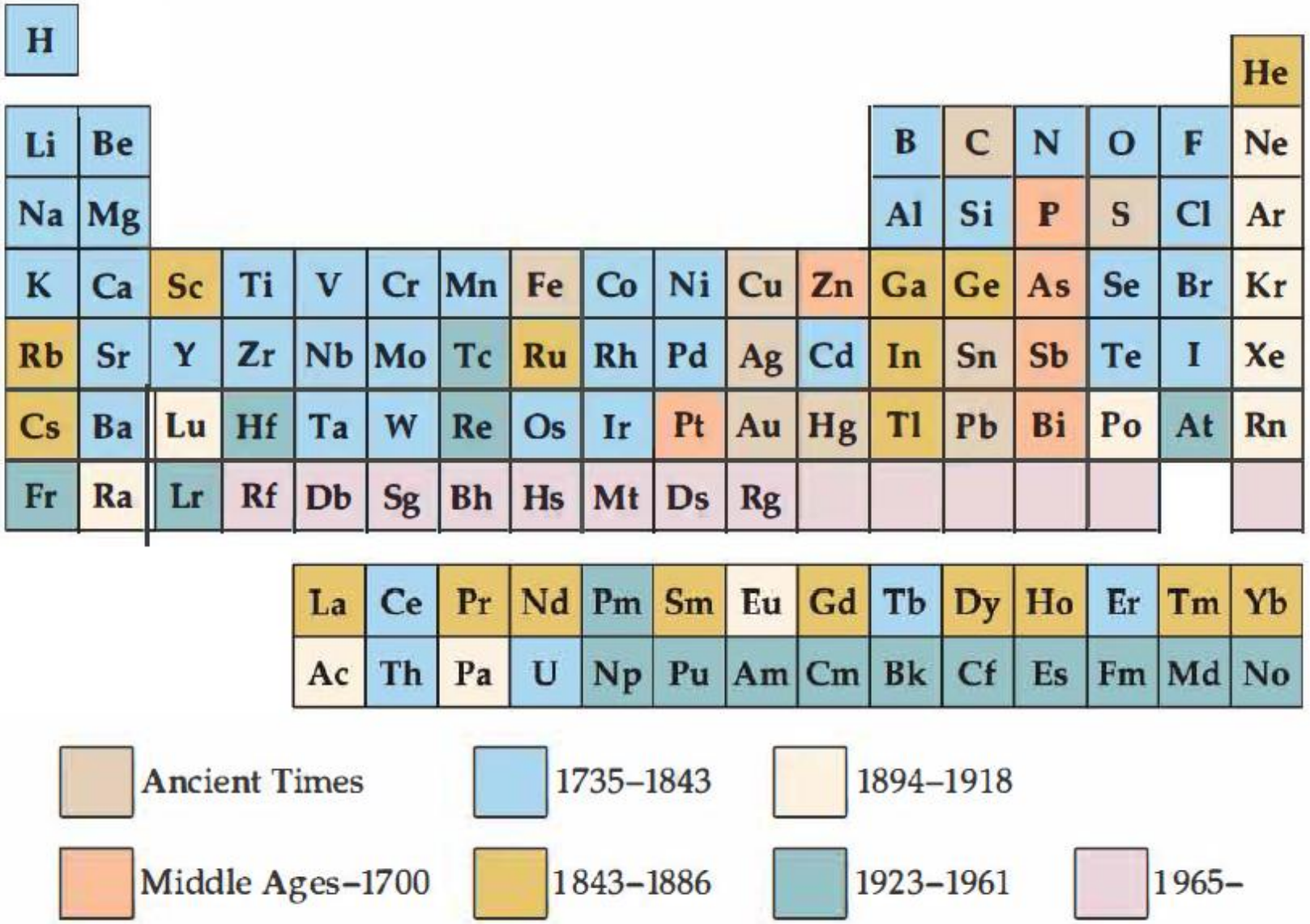
  >>> **credited for finding the periodic table**

Ge properties: Prediction *vs*. actual measurement

Property	Mendeleev's Predictions for Eka-Silicon (made in 1871)	Observed Properties of Germanium (discovered in 1886)
Atomic weight	72	72.59
Density (g/cm^3)	5.5	5.35
Specific heat (J/g-K)	0.305	0.309
Melting point (°C)	High	947
Color	Dark gray	Grayish white
Formula of oxide	XO_2	GeO_2
Density of oxide (g/cm^3)	4.7	4.70
Formula of chloride	XCl_4	$GeCl_4$
Boiling point of chloride (°C)	A little under 100	84

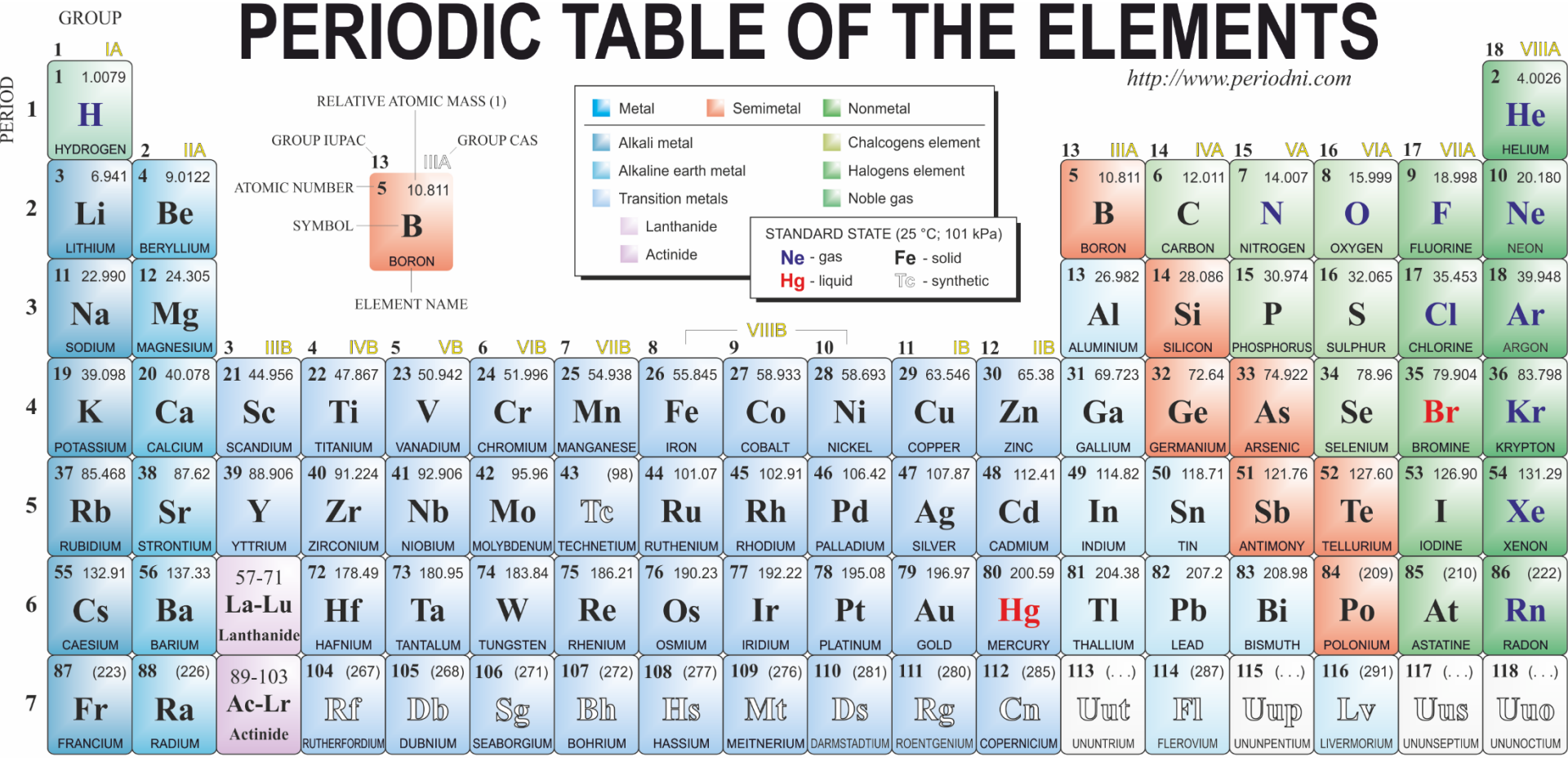


Periodic Table by Dates of Discovery



PERIODIC TABLE OF THE ELEMENTS

http://www.periodni.com



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LANTHANIDE

57	58	59	60	61	62	63	64	65	66	67	68	69	70	71
138.91	140.12	140.91	144.24	(145)	150.36	151.96	157.25	158.93	162.50	164.93	167.26	168.93	173.05	174.97
La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
LANTHANUM	CERIUM	PRASEODYMIUM	NEODYMIUM	PROMETHIUM	SAMARIUM	EUROPIUM	GADOLINIUM	TERBIUM	DYSPROSIUM	HOLMIUM	ERBIUM	THULIUM	YTTERBIUM	LUTETIUM

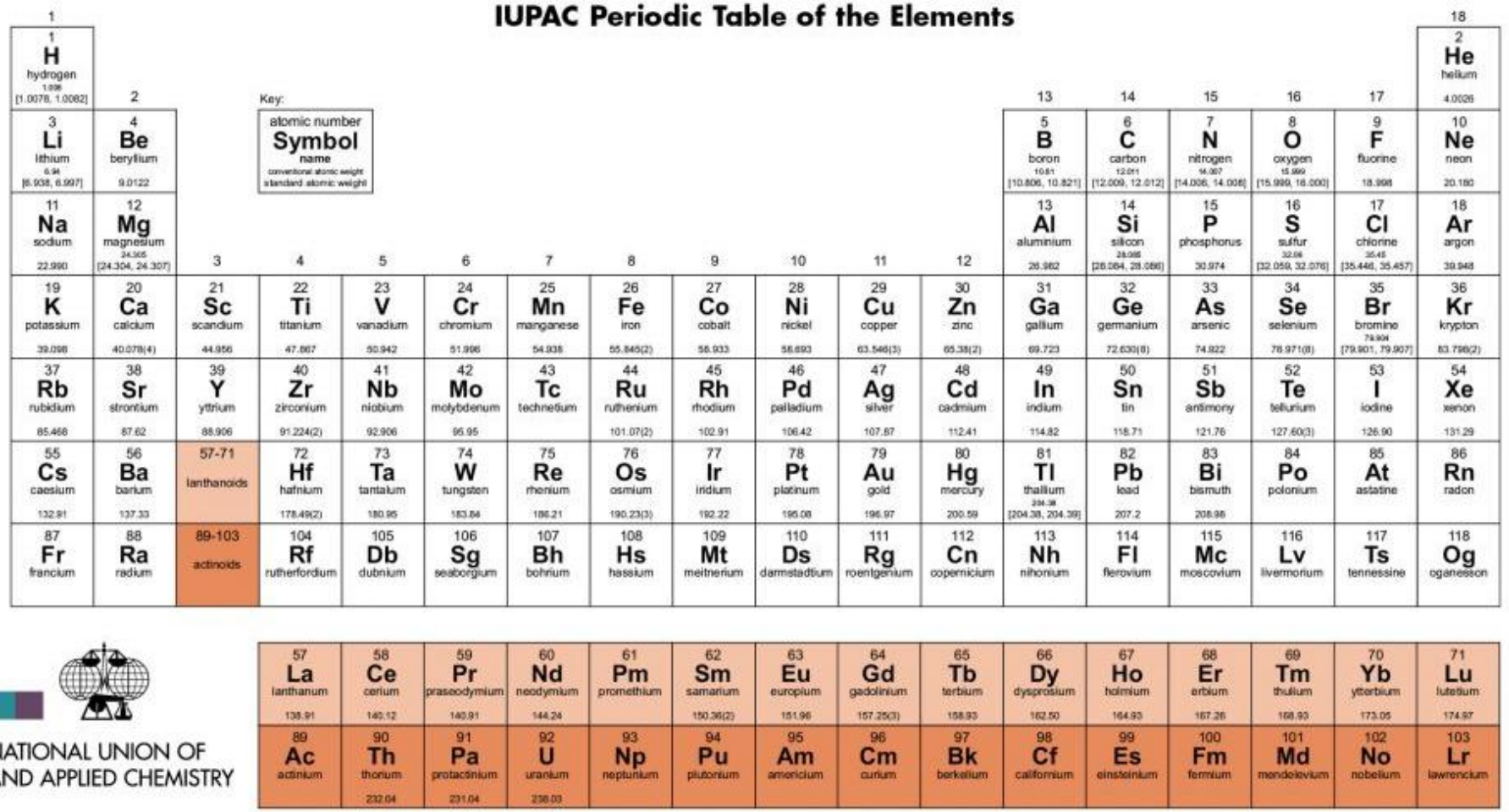
ACTINIDE

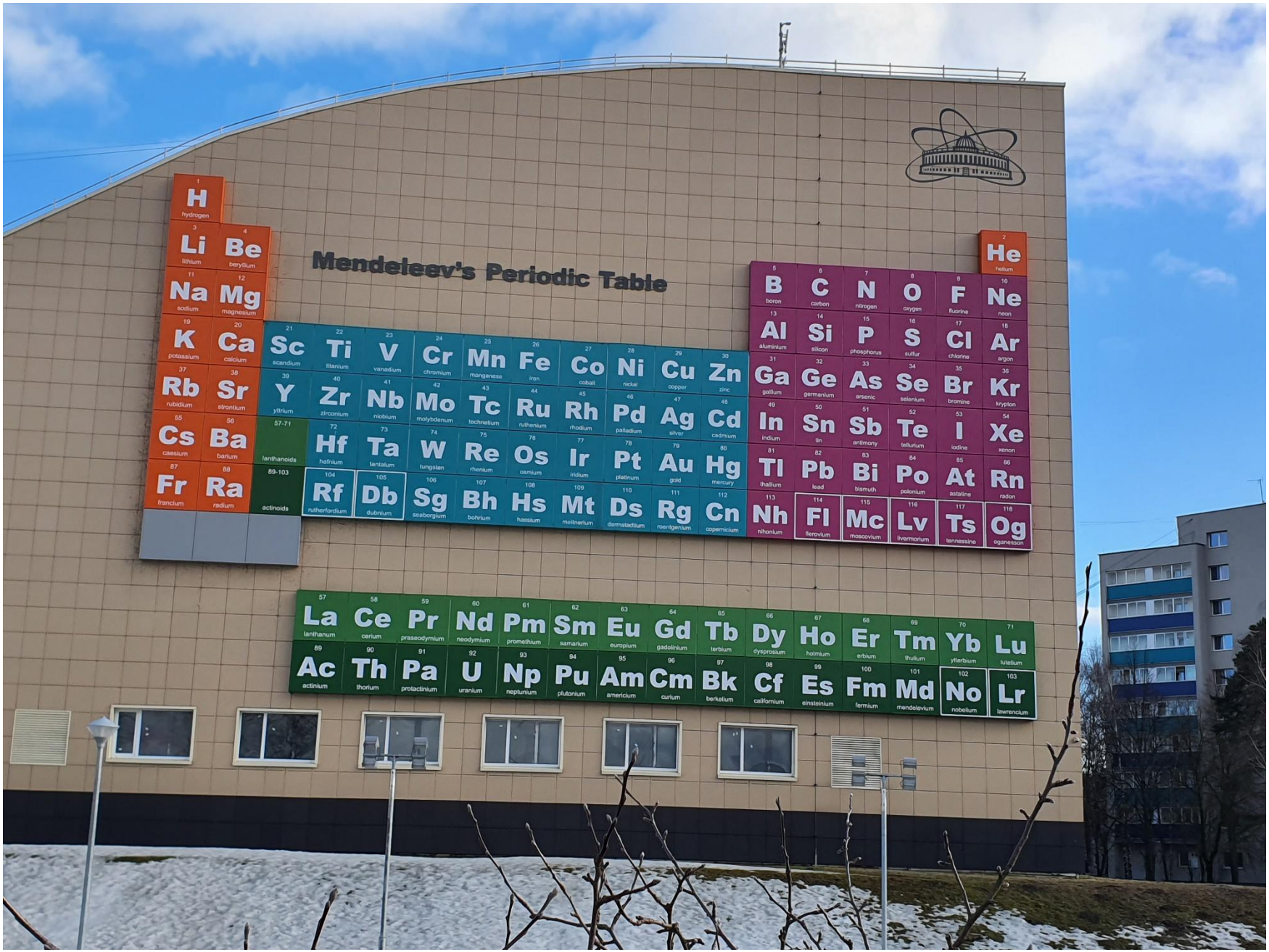
89	90	91	92	93	94	95	96	97	98	99	100	101	102	103
(227)	232.04	231.04	238.03	(237)	(244)	(243)	(247)	(247)	(251)	(252)	(257)	(258)	(259)	(262)
Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr
ACTINIUM	THORIUM	PROTACTINIUM	URANIUM	NEPTUNIUM	PLUTONIUM	AMERICIUM	CURIUM	BERKELIUM	CALIFORNIUM	EINSTEINIUM	FERMIUM	MENDELEVIUM	NOBELIUM	LAWRENCIUM

www.periodni.com

(1) Atomic Weights of the Elements 2007, Pure Appl. Chem., **81**, No. 11, 2131-2156 (2009)

The most recent version: dated Nov 28th, 2016
4 new elements 113, 115, 117 and 118





Dubna, Russia in 2022

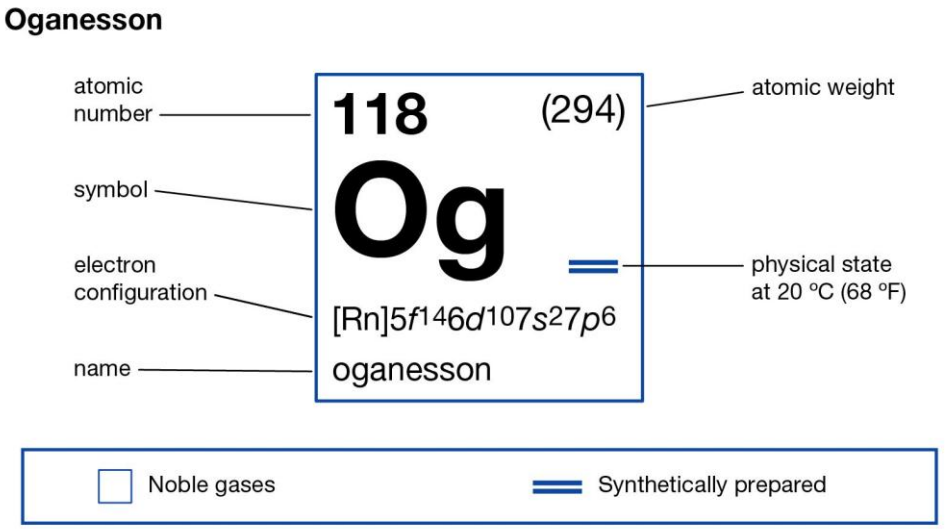


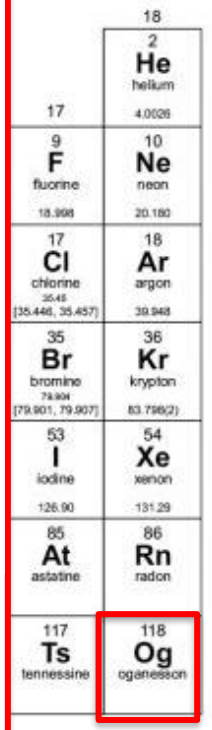
Dubna, Russia in 2022

Joint Institute for Nuclear Research

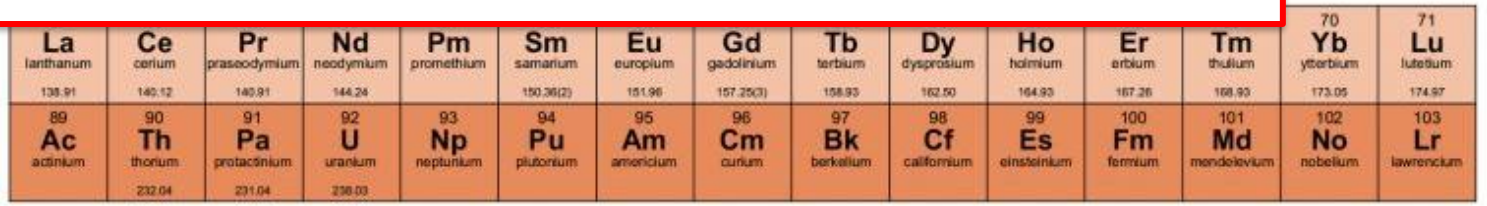


Prof. Acad. Yuri Oganessian
(April 14, 1933)
JINR, Russia









For notes and updates to this table, see www.iupac.org. This version is dated 28 November 2016.
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Project to synthesize **Unbinilium** (**eka-radium** or **element 120)**



Prof. Acad. Yuri Oganessian
(April 14, 1933)
JINR, Russia

$$^{249}_{89}\mathbf{Cf} + ^{50}_{22}\mathbf{Ti} \rightarrow ^{299}_{120}\mathbf{Ubn^*} \rightarrow \mathbf{Unb}$$

Experiment: 2025 (*expected*)

^{249}Cf target: produced by the Oak Ridge National Lab, US;

^{50}Ti beam: produced by the Hubert Curien Pluridisciplinary Institute, Strasbourg, France

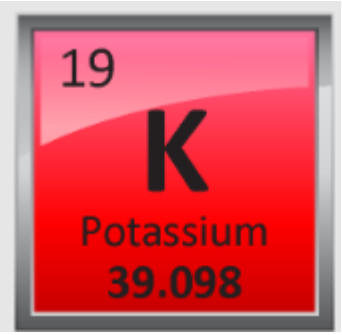
<u>Potential alternative</u>:

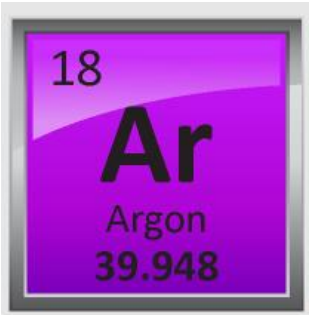
$$^{248}_{96}\mathbf{Cm} + ^{54}_{24}\mathbf{Cr} \rightarrow ^{302}_{120}\mathbf{Ubn^*} \rightarrow \mathbf{Ubn}$$

Issues with Mendeleev table

- Based on atomic mass
- Properties of some elements do not follow the proposed table
- $M_{Ar} = 19 > M_K = 18$
- Ar has similar properties as Ne and Kr

Li	Be		B	C	N	O	F	Ne
Na	Mg		Al	Si	P	S	Cl	Ar
K	Ca		Ga	Ge	As	Se	Br	Kr

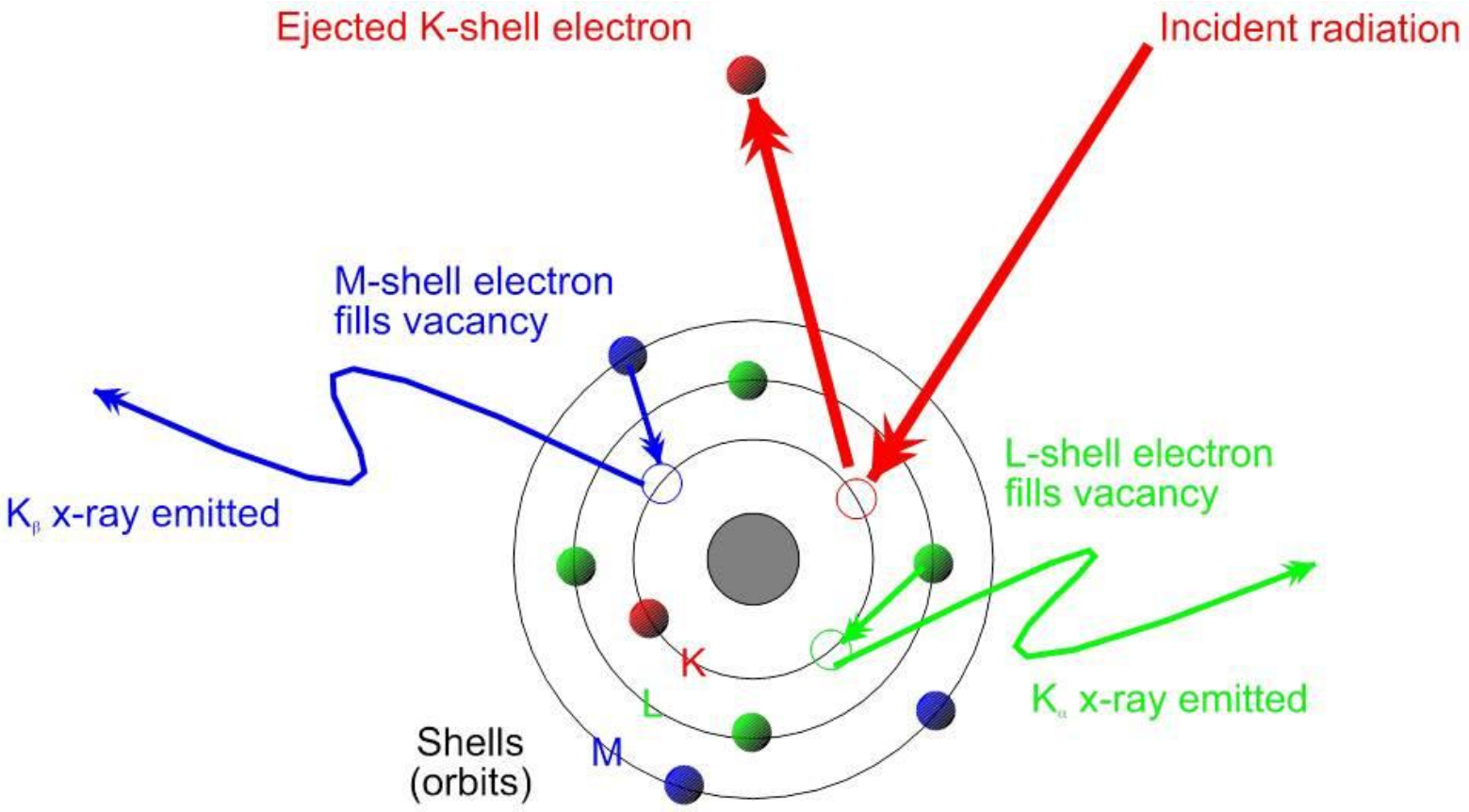




Discovery of atomic number

- In 1913, Henry Moseley (1887-1915) developed the concept of atomic number.
- Each element produces X-rays of a unique frequency
- Identified the atomic number as the number of protons in the nucleus of the atom.
- When elements are arranged in order of increasing atomic number, rather than increasing atomic weight, Ar and K appear in their correct places in the table

Atomic X ray: Characteristic for each atom



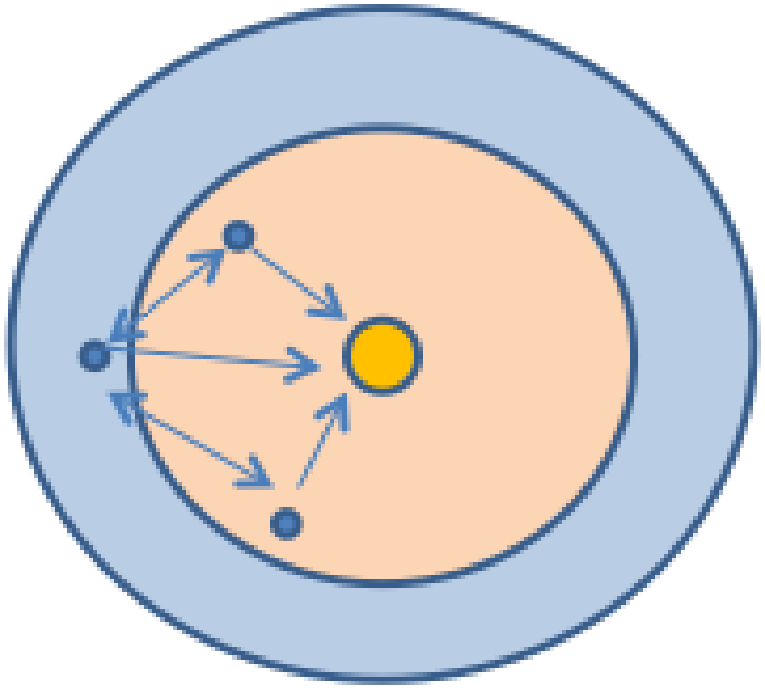
$$\nu = A \cdot (Z - b)^2$$

2. EFFECTIVE NUCLEAR CHARGE

- Many of the properties of atoms depend on their electron configurations and on how strongly their outer electrons are attracted to the nucleus.
- The force of attraction between an electron and the nucleus depends on
  - the magnitude of the net nuclear charge acting on the electron, and
  - the average distance between the nucleus and the electron

Effective nuclear charge

- **Shielding effect**: In a many-electron atom, each electron is simultaneously attracted to the nucleus and repelled by the other electrons.



Effective nuclear charge

- The net attraction of each electron to the nucleus is combination of the interaction with
  - the average environment created by the nucleus and
  - the other electrons in the atom

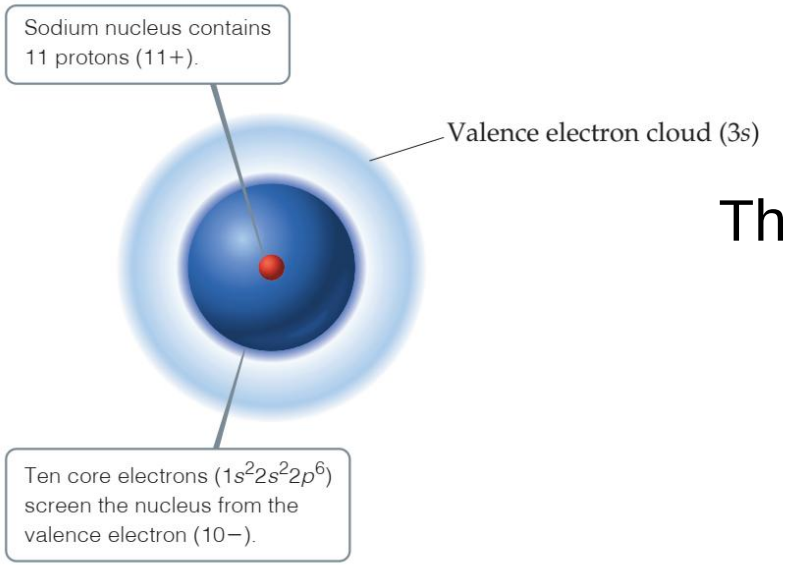
Effective nuclear charge

Effective nuclear charge (Z_{eff}): net positive charge of the nucleus experienced by the outer electrons

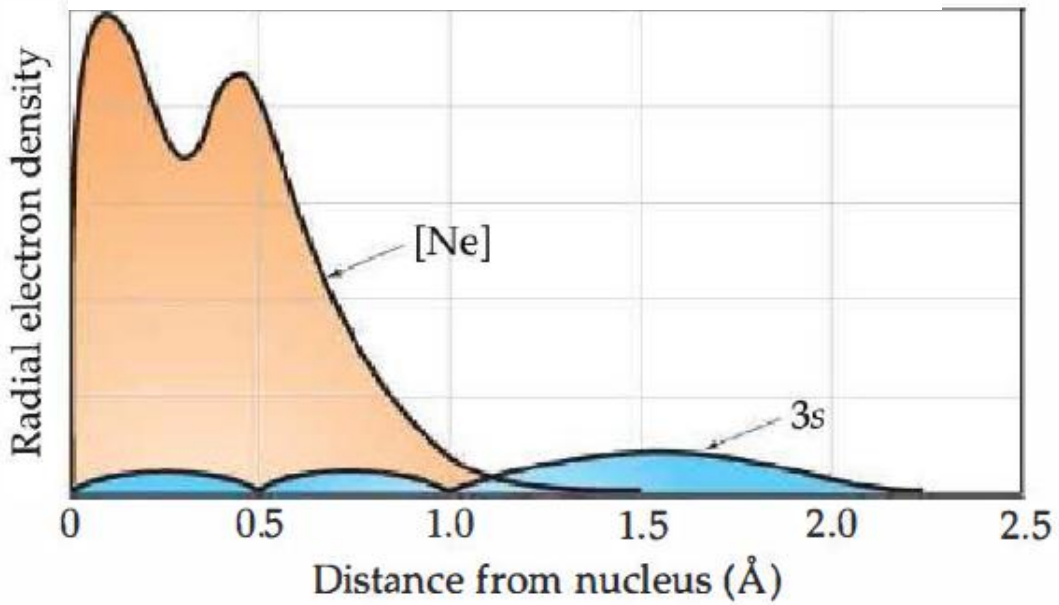
$$Z_{eff} = Z - S$$

Z_{eff} : effective nuclear charge
Z: nuclear charge = the number of proton in the nucleus
S: shielding or screening constant (greater than 0 but less than *Z*)
S is usually close to the number of core electrons in an atom

Effective nuclear charge



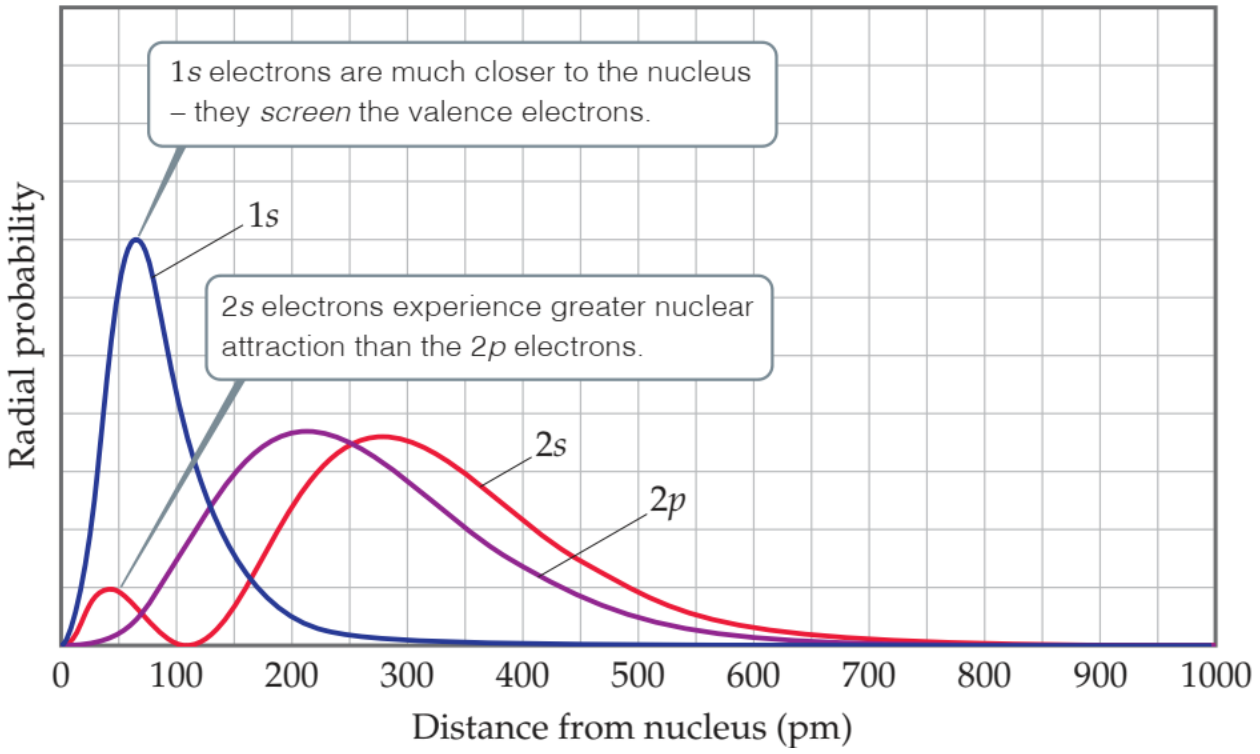
The effective nuclear charge in Na



The 3s electron has some probability of being inside the Ne core.
The core is not completely effective in screening the 3s electron from the nucleus. Thus, the effective nuclear charge experienced by the 3s electron is somewhat greater than 1+

Effective nuclear charge and electron energy

- A carbon atom with electron configuration: $1s^2 2s^2 2p^2$
- The energy of the 2p orbital (l = 1) > 2s orbital (l = 0) even though both of these orbitals are in the n = 2 shell
- An electron in the 2s orbital is less effectively screened/shielded by the core orbitals than is an electron in the 2p orbital



Effective nuclear charge

- In the periodic table, the effective nuclear charge increases as we move across any row (period) of the table

- Going down a column, the effective nuclear charge experienced by valence electrons changes far less than it does across a row – another words *“effective nuclear charge increase slightly as we go down a column”*



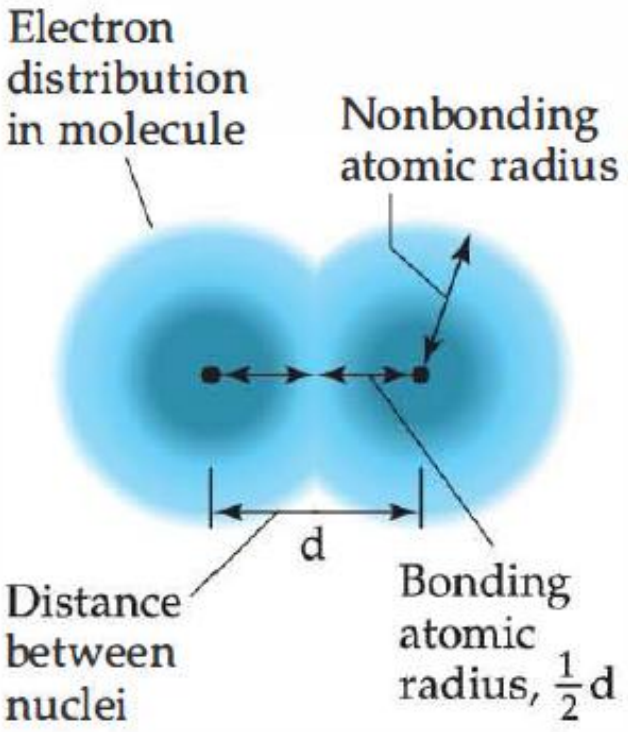


Z or Z_{eff}	H																	He
Z	1.00																	2.00
1s	1.00																	1.69
	Li	Be											B	C	N	O	F	Ne
Z	3.00	4.00											5.00	6.00	7.00	8.00	9.00	10.00
1s	2.69	3.69											4.68	5.67	6.67	7.66	8.65	9.64
2s	1.28	1.91											2.58	3.22	3.85	4.49	5.13	5.76
2p													2.42	3.14	3.83	4.45	5.10	5.76
	Na	Mg											Al	Si	P	S	Cl	Ar
Z	11.00	12.00											13.00	14.00	15.00	16.00	17.00	18.00
1s	10.63	11.61											12.59	13.58	14.56	15.54	16.52	17.51
2s	6.57	7.39											8.21	9.02	9.83	10.63	11.43	12.23
2p	6.80	7.83											8.96	9.95	10.96	11.98	12.99	14.01
3s	2.51	3.31											4.12	4.90	5.64	6.37	7.07	7.76
3p													4.07	4.29	4.89	5.48	6.12	6.76
	K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
Z	19.00	20.00	21.00	22.00	23.00	24.00	25.00	26.00	27.00	28.00	29.00	30.00	31.00	32.00	33.00	34.00	35.00	36.00
1s	18.49	19.47	20.46	21.44	22.43	23.41	24.40	25.38	26.37	27.35	28.34	29.33	30.31	31.29	32.28	33.26	34.25	35.23
2s	13.01	13.78	14.57	15.38	16.18	16.98	17.79	18.60	19.41	20.21	21.02	21.83	22.60	23.37	24.13	24.89	25.64	26.40
2p	15.03	16.04	17.06	18.07	19.07	20.08	21.08	22.09	23.09	24.10	25.10	26.10	27.09	28.08	29.07	30.07	31.06	32.05
3s	8.68	9.60	10.34	11.03	11.71	12.37	13.02	13.68	14.32	14.96	15.59	16.22	17.00	17.79	18.60	19.40	20.22	21.03
3p	7.73	8.66	9.41	10.10	10.79	11.47	12.11	12.78	13.44	14.09	14.73	15.37	16.20	17.01	17.85	18.71	19.57	20.43
4s	3.50	4.40	4.63	4.82	4.98	5.13	5.28	5.43	5.58	5.71	5.84	5.97	7.07	8.04	8.94	9.76	10.55	11.32
3d			7.12	8.14	8.98	9.76	10.53	11.18	11.86	12.53	13.20	13.88	15.09	16.25	17.38	18.48	19.56	20.63
4p													6.22	6.78	7.45	8.29	9.03	9.34
	Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
Z	37.00	38.00	39.00	40.00	41.00	42.00	43.00	44.00	45.00	46.00	47.00	48.00	49.00	50.00	51.00	52.00	53.00	54.00
1s	36.21	37.19	38.18	39.16	40.14	41.13	42.11	43.09	44.08	45.06	46.04	47.03	48.01	48.99	49.97	50.96	51.94	52.92
2s	27.16	27.90	28.62	29.37	30.13	30.88	31.63	32.38	33.16	33.88	34.63	35.39	36.12	36.86	37.60	38.33	39.07	39.80
2p	33.04	34.03	35.00	35.99	36.98	37.97	38.94	39.95	40.94	41.93	42.92	43.91	44.90	45.89	46.87	47.86	48.85	49.84
3s	21.84	22.66	23.55	24.36	25.17	25.98	26.79	27.60	28.44	29.22	30.03	30.84	31.63	32.42	33.21	34.00	34.79	35.58
3p	21.30	22.17	23.09	23.85	24.62	25.47	26.38	27.22	28.15	29.02	29.81	30.69	31.52	32.35	33.18	34.01	34.84	35.67
4s	12.39	13.44	14.26	14.90	15.28	16.10	17.20	17.66	18.58	18.99	19.87	20.87	21.76	22.66	23.54	24.41	25.30	26.17
3d	21.68	22.73	25.40	25.57	26.25	27.23	28.35	29.36	30.41	31.45	32.54	33.61	34.68	35.74	36.80	37.84	38.90	39.95
4p	10.88	11.93	12.75	13.46	14.08	14.98	15.81	16.44	17.14	17.72	18.56	19.41	20.37	21.27	22.18	23.12	24.03	24.96
5s	4.99	6.07	6.26	6.45	5.92	6.11	7.23	6.49	6.64	(empty)	6.76	8.19	9.51	10.63	11.62	12.54	13.40	14.22
4d			15.96	13.07	11.24	11.39	12.88	12.81	13.44	13.62	14.76	15.88	16.94	17.97	18.97	19.96	20.93	21.89
5p													8.47	9.10	10.00	10.81	11.61	12.43

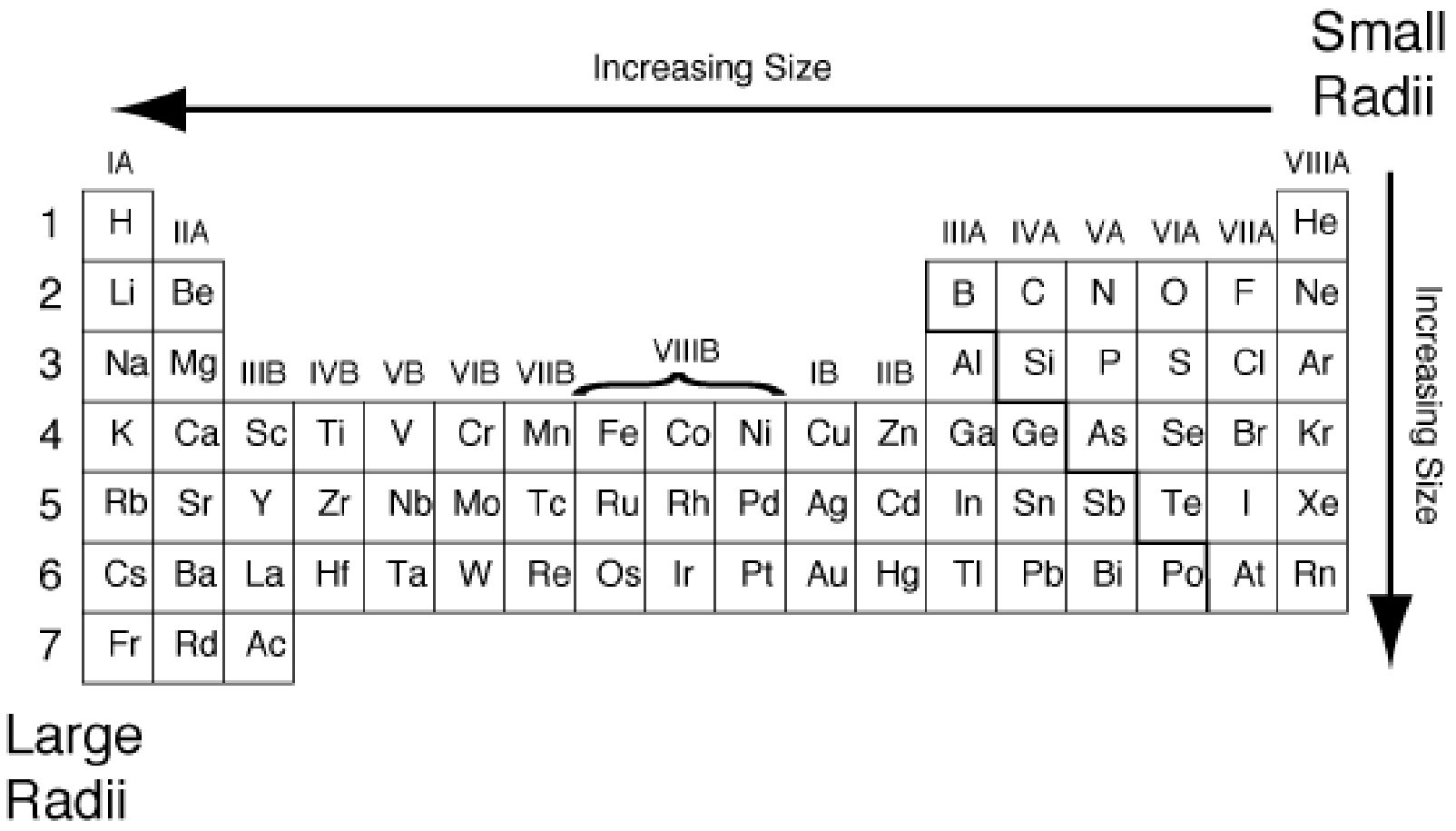
3. SIZES OF ATOMS AND IONS

- Atoms and ions do not have sharply defined boundaries at which the electron distribution becomes zero.

Size of atom: The closest distances separating the nuclei during such collisions determine the apparent radii of the atoms

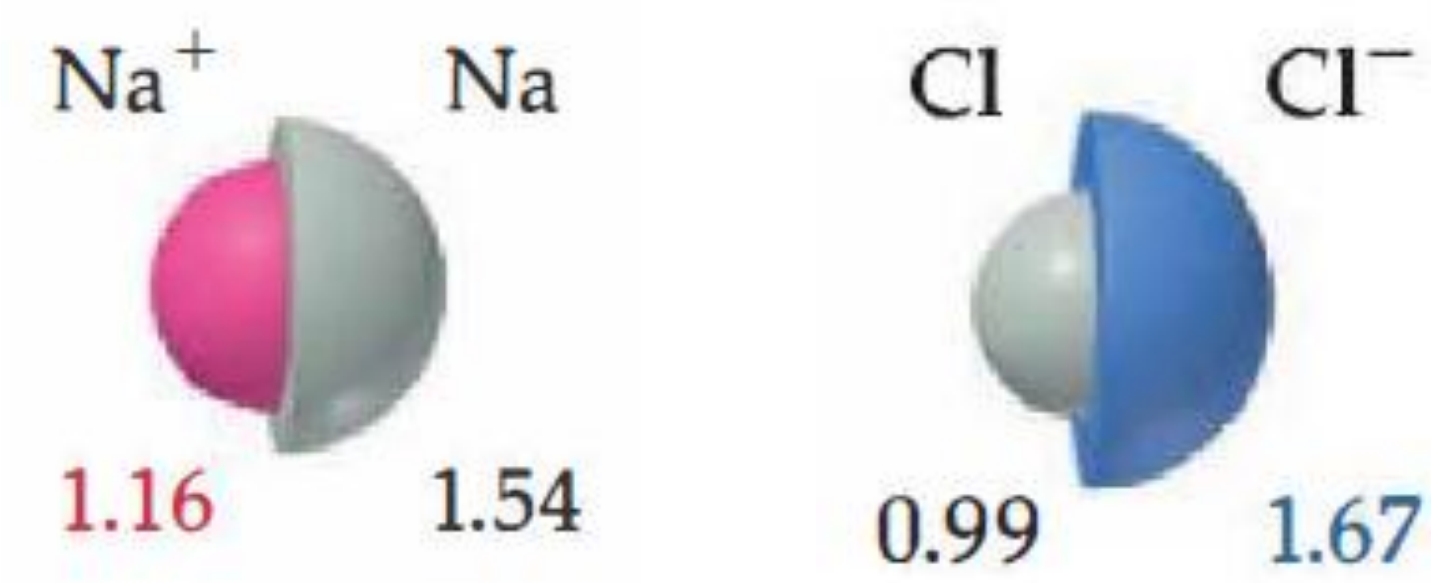


Periodic Trends in Atomic Radii

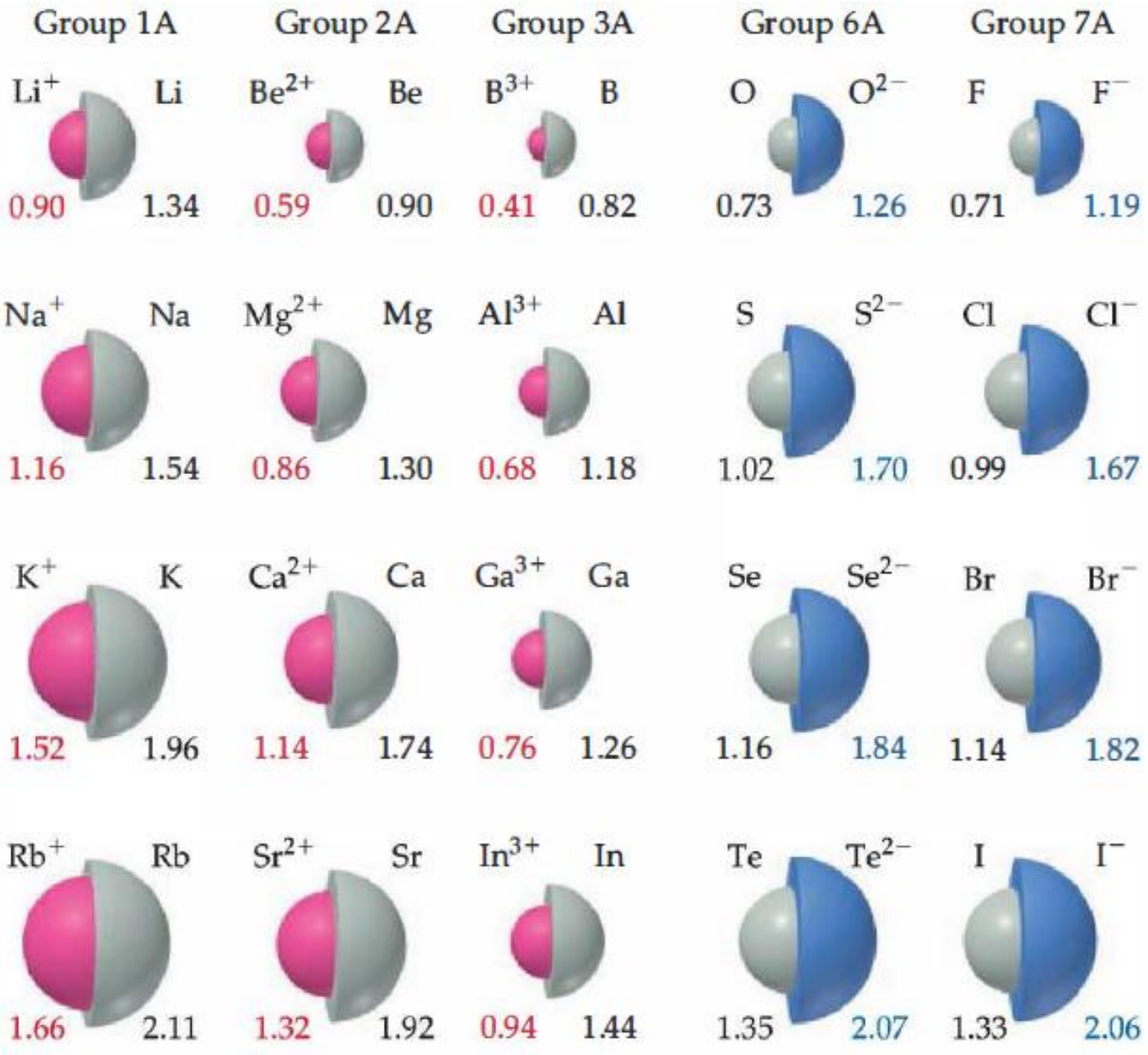


Periodic Trends in Ionic Radii

- Cations are smaller than their parent atoms
- Anions are larger than their parent atoms



Periodic Trends in Ionic Radii



For ions carrying the same charge, size increases as we move down a column in the periodic table.

Isoelectronic series

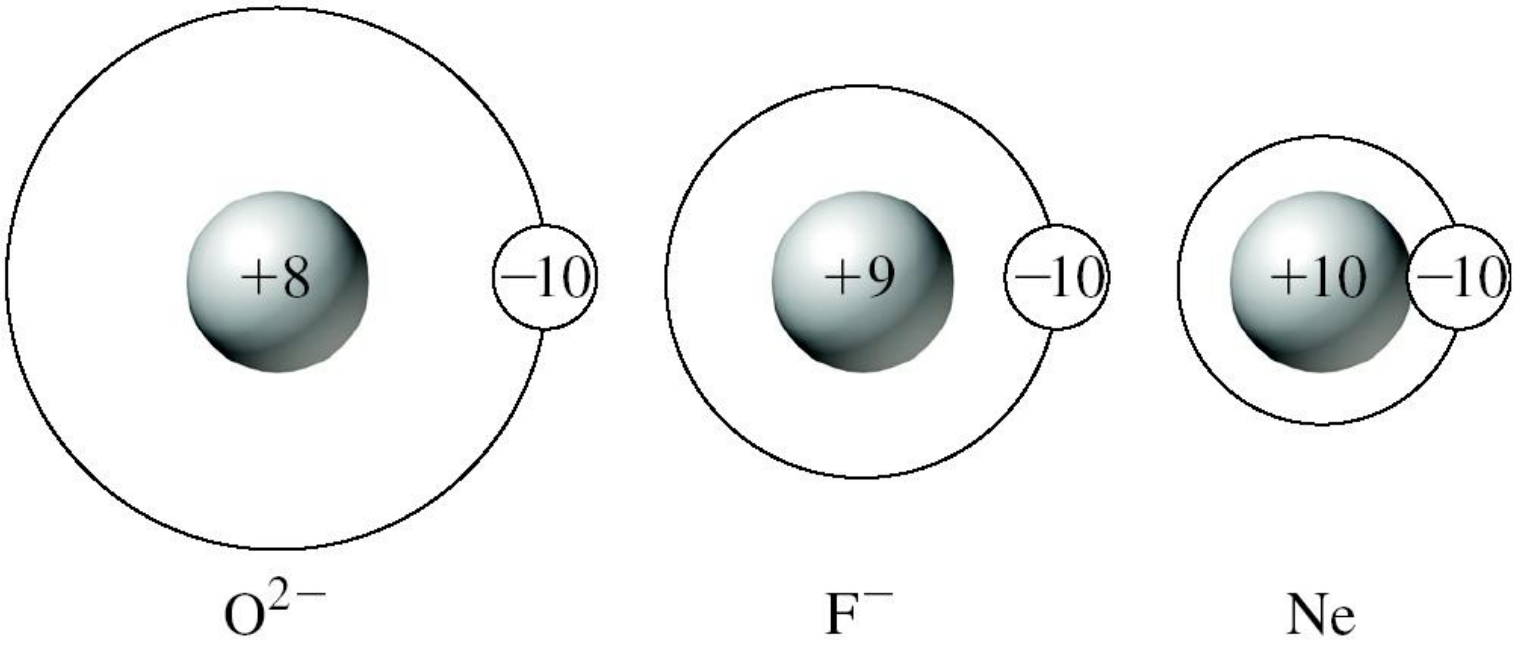
- Group of ions all containing the same number of electrons
- O^{2-}, F^{-}, Na^{+}, Mg^{2+}, Al^{3+}
- Each has 10 electrons

Isoelectronic series

—Increasing nuclear charge ⟶

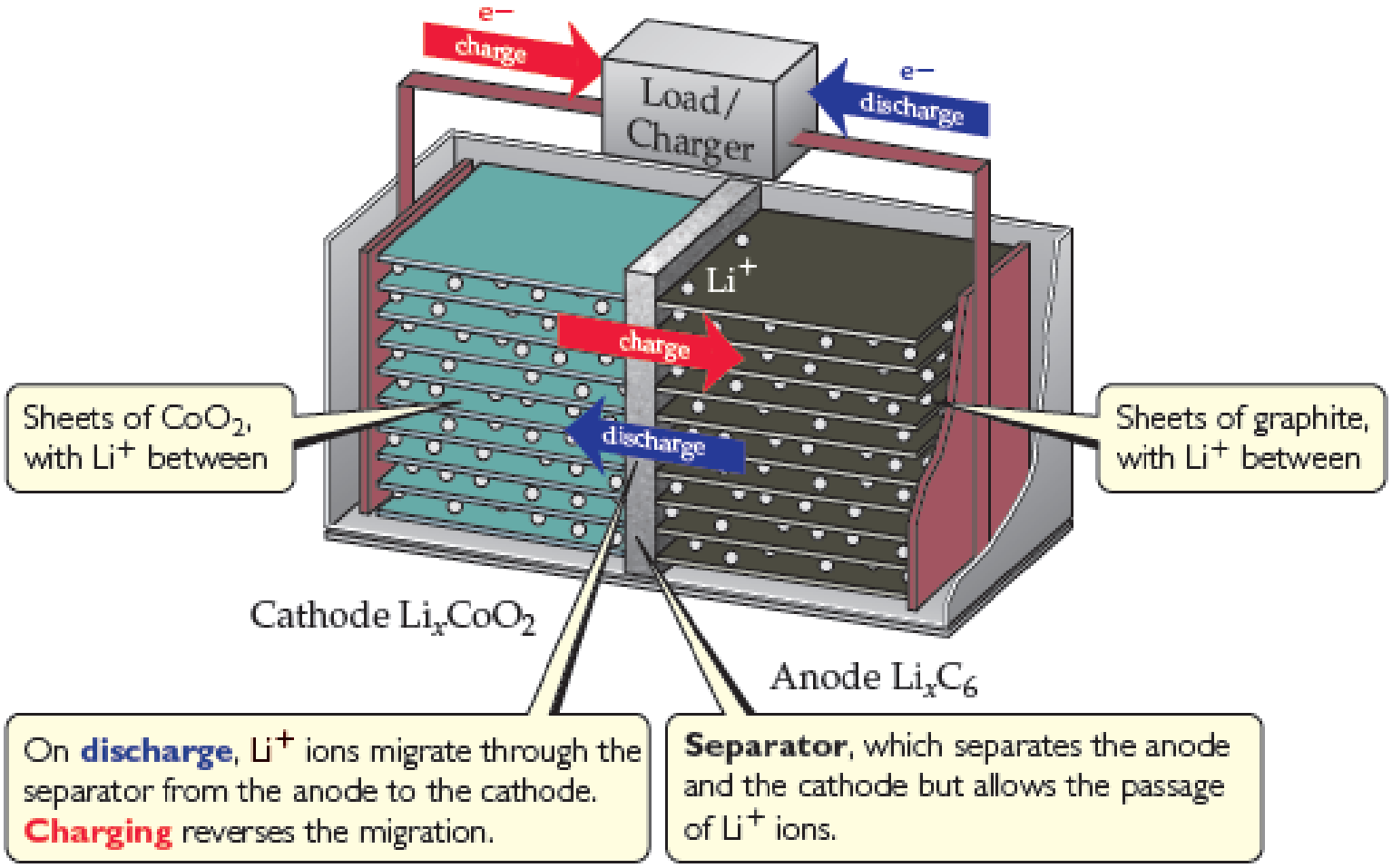
O^{2-}	F^-	Na^+	Mg^{2+}	Al^{3+}
1.26 Å	1.19 Å	1.16 Å	0.86 Å	0.68 Å

—Decreasing ionic radius ⟶



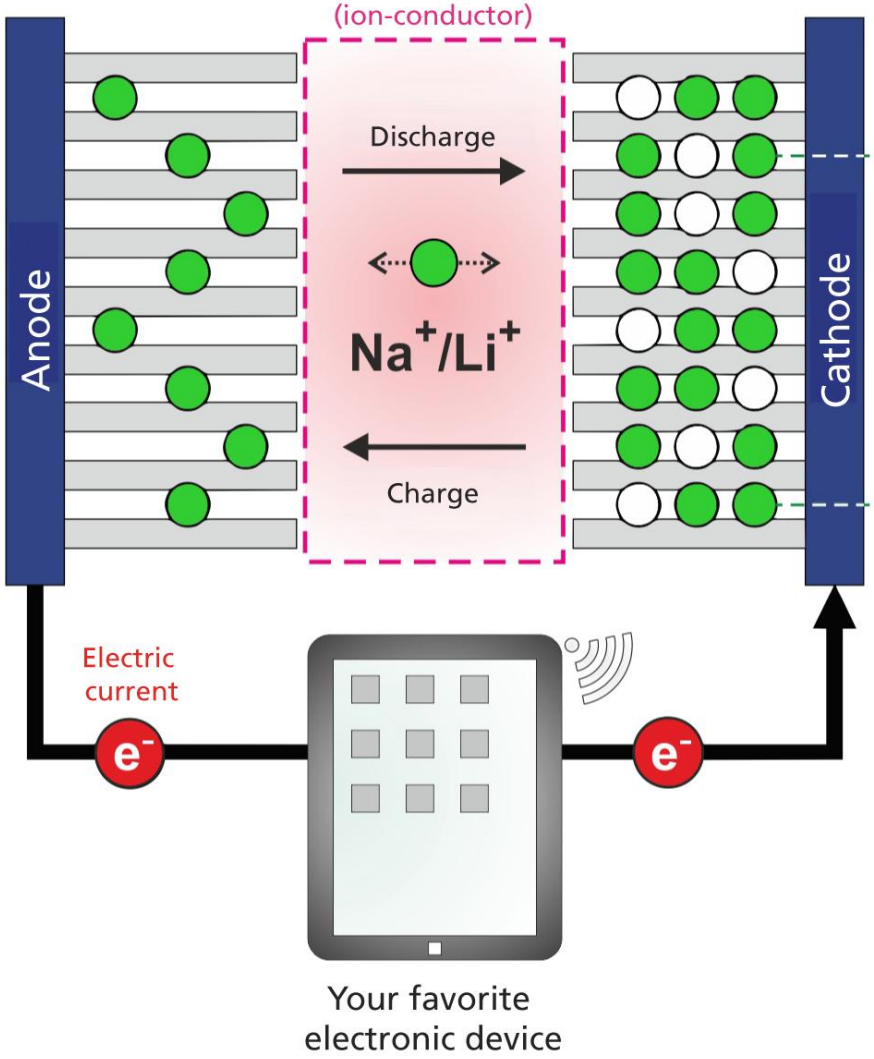
Is ion size important in devices engineering?

The case of Li ion battery:

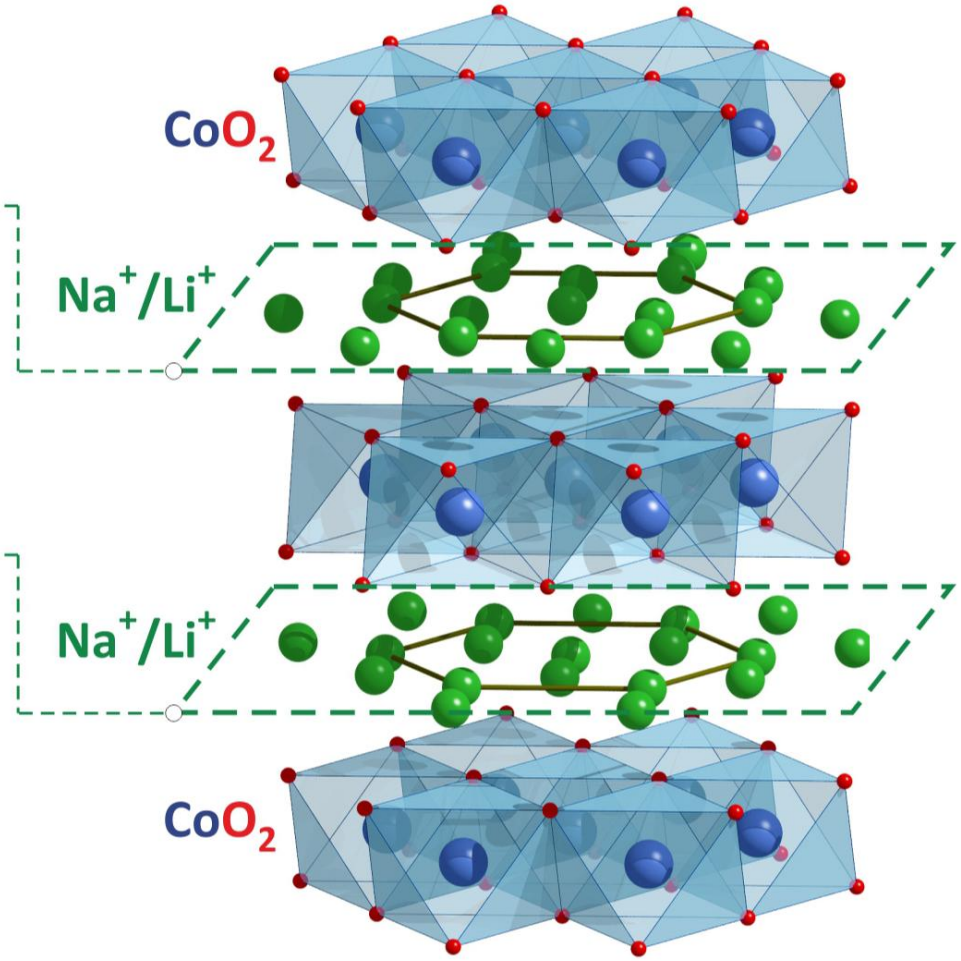


▲ **Figure 7.9** **Schematic of a lithium-ion battery.**

Working Principle for a Rechargeable Battery



Electrode Material $LiCoO_2$ / $NaCoO_2$



Arrange following ions in order of increasing ionic radius :

O^{2-}, F^{-}, Na^{+}, Mg^{2+}, Al^{3+}

a) $O^{2-} < F^{-} < Na^{+} < Mg^{2+} < Al^{3+}$

b) $Na^{+} < Mg^{2+} < Al^{3+} < O^{2-} < F^{-}$

c) $Al^{3+} < Mg^{2+} < Na^{+} < F^{-} < O^{2-}$

d) $Al^{3+} < Mg^{2+} < Na^{+} < O^{2-} < F^{-}$

Arrange following ions in order of increasing ionic radius :

O^{2-}, F^{-}, Na^{+}, Mg^{2+}, Al^{3+}

a) $O^{2-} < F^{-} < Na^{+} < Mg^{2+} < Al^{3+}$

b) $Na^{+} < Mg^{2+} < Al^{3+} < O^{2-} < F^{-}$

c) $\mathbf{Al^{3+} < Mg^{2+} < Na^{+} < F^{-} < O^{2-}}$

d) $Al^{3+} < Mg^{2+} < Na^{+} < O^{2-} < F^{-}$

Arrange the following atoms and ions in order of increasing ionic radius : F, S^{2-}, Cl, Se^{2-}

a) F < S^{2-} < Cl < Se^{2-}

b) F < Cl < S^{2-} < Se^{2-}

c) F < S^{2-} < Se^{2-} < Cl

d) Cl < F < Se^{2-} < S^{2-}

e) S^{2-} < F < Se^{2-} < Cl

Arrange the following atoms and ions in order of increasing ionic radius : F, S^{2-}, Cl, Se^{2-}

a) F < S^{2-} < Cl < Se^{2-}

b) F < Cl < S^{2-} < Se^{2-}

c) F < S^{2-} < Se^{2-} < Cl

d) Cl < F < Se^{2-} < S^{2-}

e) S^{2-} < F < Se^{2-} < Cl

Arrange following Hal_2 compounds in order of increasing Hal – Hal bond length: F_2, Cl_2, Br_2, I_2

A) $I_2 < Br_2 < Cl_2 < F_2$

B) $F_2 < Cl_2 < Br_2 < I_2$

C) $I_2 \sim Br_2 < Cl_2 \sim F_2$

D) $F_2 \sim Cl_2 < Br_2 \sim I_2$

4. IONIZATION ENERGY

- Ionization energy of an atom or ion is the minimum energy required to remove an electron from the ground state of the isolated gaseous atom or ion

IONIZATION ENERGY

- First ionization energy, I_1, needed to remove the first electron from a neutral atom

$$Na(g) \rightarrow Na^+(g) + e^-$$

- Second ionization energy, I_2, is the energy needed to remove the second electron

$$Na^+(g) \rightarrow Na^{2+}(g) + e^-$$

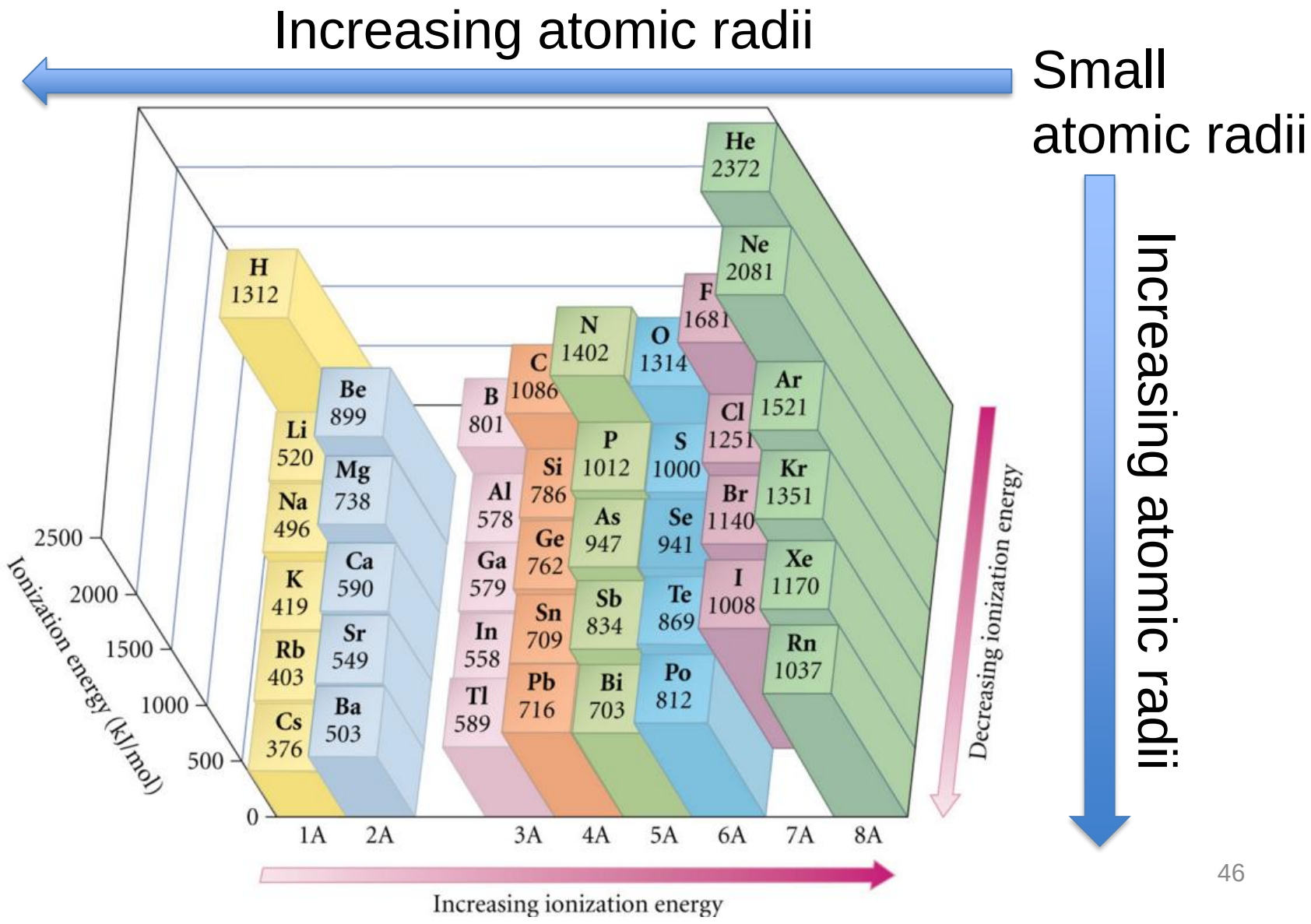
Variations in Successive Ionization Energies

TABLE 7.2 ■ Successive Values of Ionization Energies, *I*, for the Elements Sodium through Argon (kJ/mol)

Element	I_1	I_2	I_3	I_4	I_5	I_6	I_7
Na	495	4562	(inner-shell electrons)				
Mg	738	1451	7733				
Al	578	1817	2745	11,577			
Si	786	1577	3232	4356	16,091		
P	1012	1907	2914	4964	6274	21,267	
S	1000	2252	3357	4556	7004	8496	27,107
Cl	1251	2298	3822	5159	6542	9362	11,018
Ar	1521	2666	3931	5771	7238	8781	11,995

This trend exists because with each successive removal, an electron is being pulled away from an increasingly more positive ion, requiring increasingly more energy

Periodic Trends in First Ionization Energies



Which of the following chemical equations is connected to the definitions of the second ionization energy of oxygen

a) $O_{(g)} + e^- \rightarrow O^-_{(g)}$

b) $O_{(g)} \rightarrow O^+_{(g)} + e^-$

c) $O_{(g)} + 2e^- \rightarrow O^{2-}_{(g)}$

d) $O_{(g)} \rightarrow O^{2+}_{(g)} + 2e^-$

e) $O^+_{(g)} \rightarrow O^{2+}_{(g)} + e^-$

Which of the following chemical equations is connected to the definitions of the second ionization energy of oxygen

a) $O_{(g)} + e^- \rightarrow O^-_{(g)}$

b) $O_{(g)} \rightarrow O^+_{(g)} + e^-$

c) $O_{(g)} + 2e^- \rightarrow O^{2-}_{(g)}$

d) $O_{(g)} \rightarrow O^{2+}_{(g)} + 2e^-$

e) $\mathbf{O^+_{(g)} \rightarrow O^{2+}_{(g)} + e^-}$

Li has a larger first ionization energy in comparison to that of Na because:

A) Nucleus of Li has more proton

B) Li atomic size is larger

C) Li is extremely sensitive to O_2 and H_2O

D) 3s electron has higher potential energy than the 2s electron

Electron affinities

- Electron affinity is the energy change that occurs when an electron is added to a gaseous atom

Ionization energy vs. electron affinity

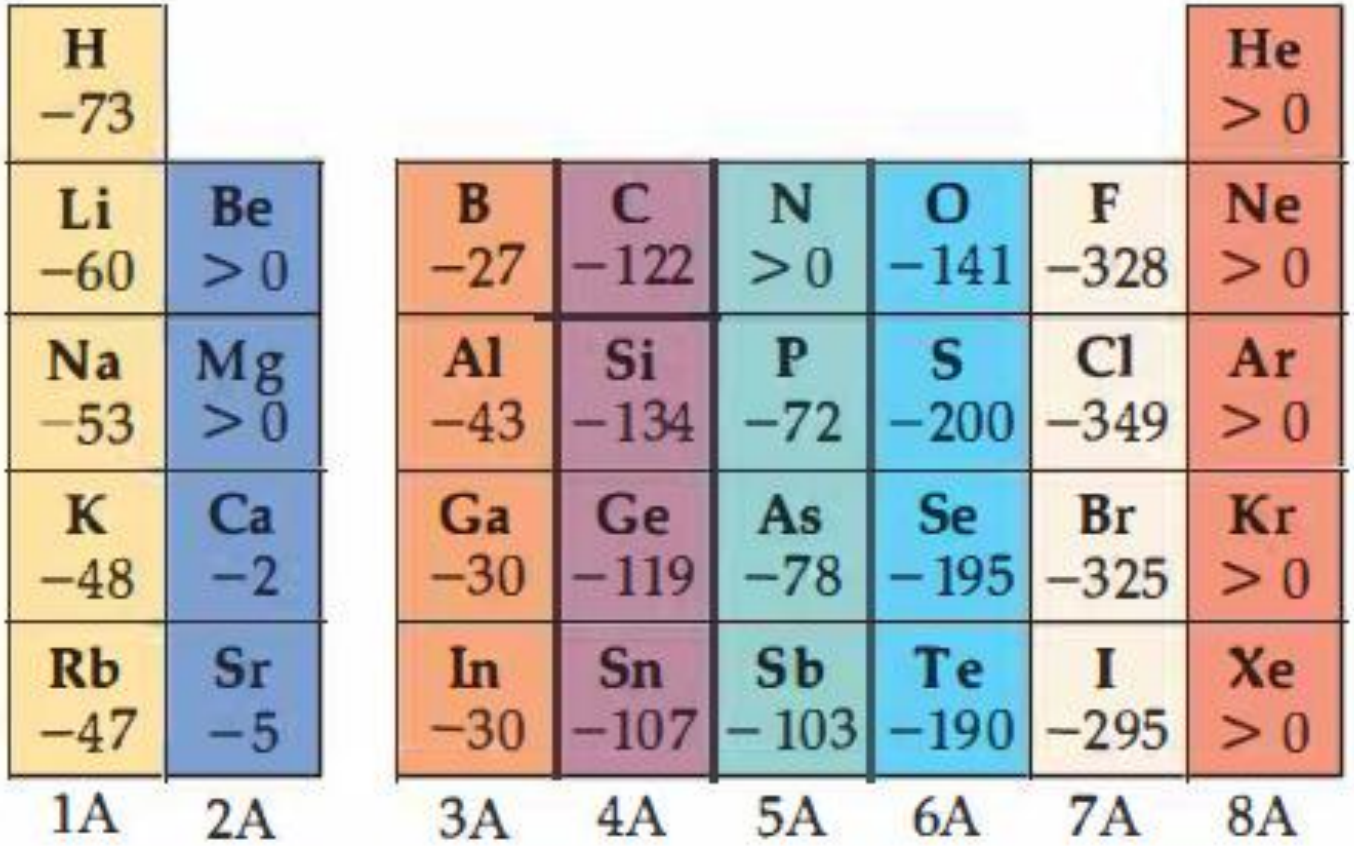
- Ionization energy: how easy an atom loses an electron

$$\textit{Ionization energy:}\ \underset{[Ne]3s^23p^5}{Cl(g)} \longrightarrow \underset{[Ne]3s^23p^4}{Cl^+(g)} + e^- \qquad \Delta E = 1251\ \text{kJ/mol}$$

- Electron affinity: how easy an atom gains an electron.

$$\textit{Electron affinity:}\ \underset{[Ne]3s^23p^5}{Cl(g)} + e^- \longrightarrow \underset{[Ne]3s^23p^6}{Cl^-(g)} \qquad \Delta E = -349\ \text{kJ/mol}$$

Electron affinity (kJ/mol)

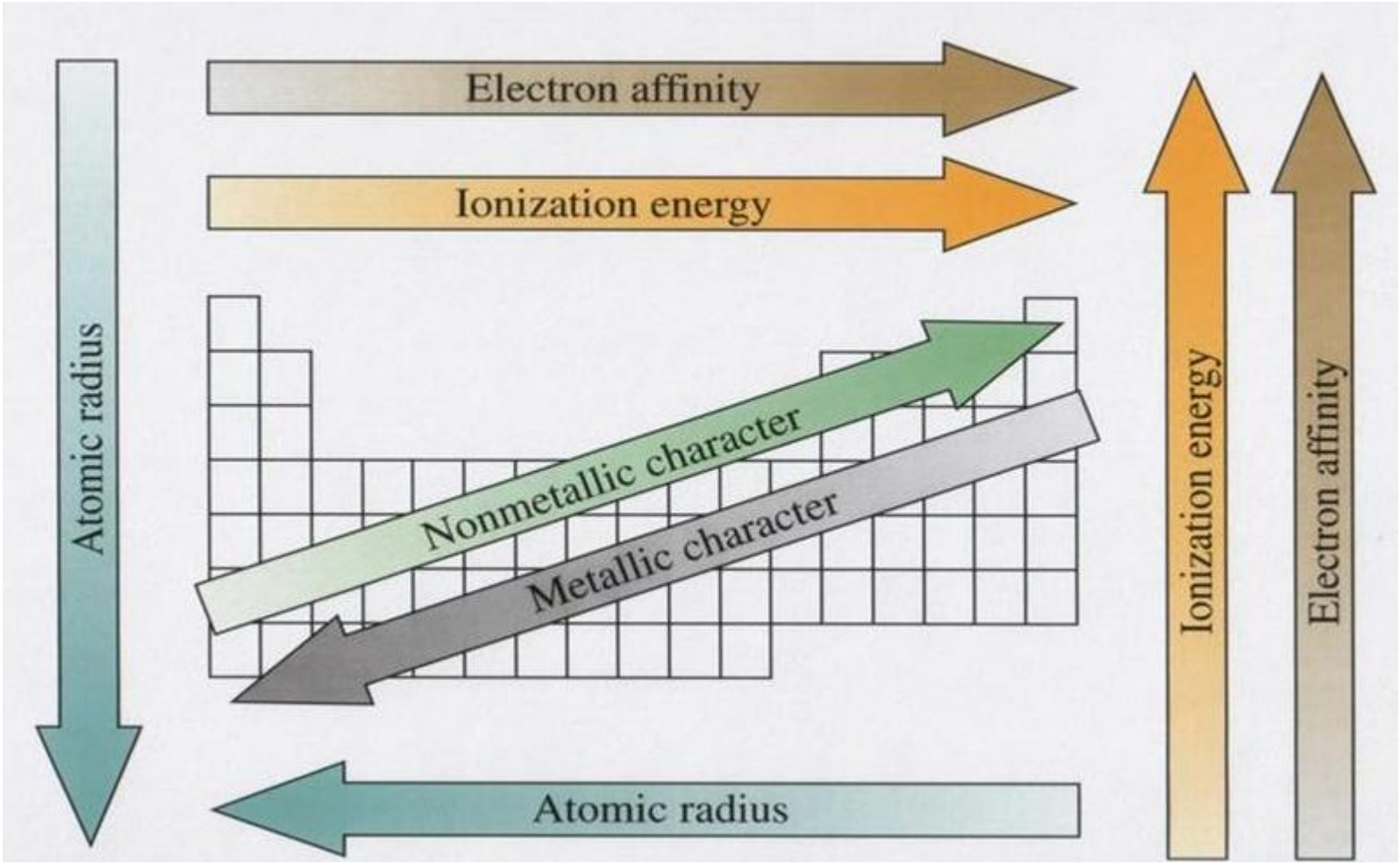


No obvious trend

Key points

- The periodic table development
- Effective nuclear charge
- Atom size, ion size
- Ionization energy
- Electron affinity

Key points


Electron affinity
Ionization energy
Atomic radius
Nonmetallic character
Metallic character
Atomic radius
Ionization energy
Electron affinity

General Chemistry I

Chemistry: The central science

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Lectures 7,8: Periodic properties of elements